Root Security Audit & Remediation (root_audit.sh)

Application Overview and Objectives

The `root_audit.sh` utility is a security orchestration script for RHEL 9+ or Ubuntu 24.04+ environments. Its primary objective is to enforce a rigorous security posture for the `root` account, aligning with modern CIS and STIG compliance requirements.

The application serves two primary functions:

1. **Security Audit**: A non-intrusive, read-only assessment of the root account's current state (password status, lock state, SSH accessibility, and cryptographic hash strength).
2. **Automated Remediation**: A deterministic process to harden the root account by enforcing complex 32-character passwords, locking the account to prevent direct interactive login, and disabling direct SSH access, all while maintaining strict operational safety.

Architecture and Design Choices

1. Modular Functional Design

The script is architected as a collection of discrete, idempotent functions. This high-level modularity includes:

- **Service Isolation**: Dedicated functions for service management (e.g., `reload_ssh_service`).
- **Audit Orchestration**: Centralized `run_full_audit` for consistent state checking.
- **Security Logic**: Decoupled helpers for sudoers path discovery and group membership detection.
- **Testability**: Functions can be sourced and unit-tested in isolation without triggering the `main` execution flow.

2. Operational Safety & Atomicity

Security tooling that modifies core system files like `/etc/shadow` or `/etc/sshd_config` must be fail-safe.

- **Atomic Edits**: Modifications to configuration files use a `mktemp` -> `sed` -> `mv` pattern. The `mv` command ensures an atomic replacement, preventing system files from being left in a corrupted state if the script is interrupted.
- **Automatic Backups**: Every file modification is preceded by a timestamped backup (`file.YYYYMMDDHHMMSS`) with preserved attributes (`cp -p`).
- **Immutable Attribute Handling**: The script detects and handles the Linux `immutable` (`i`) attribute using `lsattr` and `chattr`, ensuring it can operate on systems hardened with filesystem-level locks.

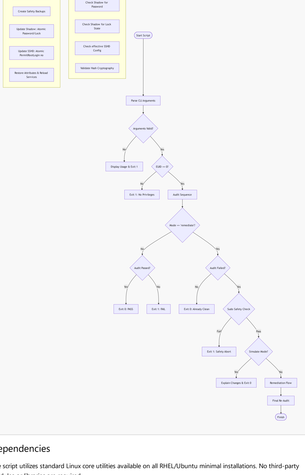
3. Non-Interactive Orchestration

Designed for execution via automation platforms (Ansible, Puppet, Salt) or service accounts, the script is fully non-interactive. It relies on deterministic CLI arguments and returns standard exit codes (0 for success/pass, non-zero for failure).

4. Sudo Safety "Kill-Switch"

A critical architectural safety feature is the `verify_sudo_users` function. It prevents administrative "brick" scenarios by aborting any destructive remediation (like locking root) if it cannot confirm the existence of at least one other non-root user with `sudo` privileges.

Data Flow and Control Logic



Dependencies

The script utilizes standard Linux core utilities available on all RHEL/Ubuntu minimal installations. No third-party modules or libraries are required.

- **Shell**: `bash` (4.0+)
- **Core Utils**: `grep`, `sed`, `awk`, `cut`, `tr`, `find`, `mktemp`, `cp`, `mv`, `date`.
- **User Management**: `passwd`, `chpasswd`, `getent`.
- **System Management**: `systemctl` (for SSHD reload).
- **Security**: `lsattr`, `chattr` (for immutable flag handling), `sshd` (for `-T` config auditing).

Command Line Arguments

Argument	Type	Default	Description
`--mode`	String	`audit`	Execution mode. Options: `audit` (read-only) or `remediate` (fix).
`--password`	String	N/A	(Remediate only) Sets the root password. **Security Note**: Prefer using the `ROOT_PASSWORD` environment variable to avoid process list visibility.
`--generate`	Flag	`false`	(Remediate only) Automatically generates a secure 32-char password.
`simulate`	Flag	`false`	(Remediate only) Explains changes without applying them (Dry-run).
`--json`	Flag	`false`	Enables CI/CD friendly JSON output on STDOUT.

Configuration via Environment Variables

For advanced orchestration, the following variables can be set to override defaults:

- `SHADOW_FILE`: Path to shadow file (default: `/etc/shadow`)
- `SSH_CONFIG`: Path to sshd_config (default: `/etc/ssh/sshd_config`)
- `SUDOERS_FILE`: Path to main sudoers (default: `/etc/sudoers`)
- `MIN_PASS_LEN`: Minimum password length policy (default: 32)

Usage Examples

1. Standard Audit

Performs a read-only security check.

```
sudo ./root_audit.sh --mode audit
```

Note: Executing the script without arguments will display the usage information.

2. Automated Remediation with Generation

Generates a 32-character password, locks the account, and hardens SSH.

```
sudo ./root_audit.sh --mode remediate --generate
```

3. Manual Remediation

Uses a pre-defined password that meets the 32-character policy.

```
sudo ./root_audit.sh --mode remediate --password
'v@ryStr0ngP@ssw0rd_32_Chars_Long!!'
```

4. Integration via Environment Overrides

Used for testing or non-standard filesystem layouts.

```
SSH_CONFIG="/custom/sshd_config" sudo ./root_audit.sh --mode audit
```

Unit Tests

The `root_audit_test.sh` suite provides comprehensive validation of the application's logic, safety mechanisms, and error handling. It is designed to run in a fully isolated environment without modifying the host system.

Test Execution

To execute the test suite, run the following command from the source directory:

```
./root_audit_test.sh
```

Note: The test suite creates a temporary workspace in `$TMPDIR/unitests/$uid` and performs an automatic cleanup upon completion.

Test Coverage Scenarios

The suite covers **28 distinct validation points**, categorized as follows:

Category	Scenarios Covered
Password Policy	Exhaustive validation of length and complexity (uppercase, lowercase, digits, special characters) including rejection paths.
Attribute Handling	Detection, temporary removal, and restoration of the `immutable` (`i`) flag during file operations.
File Safety	Verification of timestamped backups (`.YYYYMMDDHHMMSS`) and atomic state transitions.
Audit Engine	Logic validation for detecting unlocked accounts, weak cryptographic hashes (MD5), and insecure SSH configurations.
Sudo Safety Kill-Switch	Multi-path discovery simulation for administrators (groups and explicit entries) to prevent administrative lockout.
CLI Orchestration	Comprehensive exit code validation for all modes, no-argument guards, and environment variable (`ROOT_PASSWORD`) injection.
JSON Mode	Verification of schema integrity and machine-readable output for CI/CD integration.
Privilege Logic	Strict enforcement of `EUID=0` requirements for both successful execution and graceful failure.

5. Compliance Verification (Audit + Generation)

Runs a full compliance remediation with an auto-generated password and ensures all subsequent checks pass.

```
# Generate password, harden SSH, and perform a post-remediation audit
sudo ./root_audit.sh --mode remediate --generate
```

6. Secure Remediation (Environment Variable)

To avoid sensitive data appearing in process lists (`ps`) or shell history, use the `ROOT_PASSWORD` environment variable.

```
# Pass password securely via environment
export ROOT_PASSWORD='v@ryStr0ngP@ssw0rd_32_Chars_Long!!'
sudo -E ./root_audit.sh --mode remediate
```

Note: The `-E` flag for `sudo` is used to preserve the environment variable.

Remote Execution via SSH

When executing `root_audit.sh` remotely via SSH, you can pass the `ROOT_PASSWORD` variable securely using the following patterns.

1. Direct Command Prefix (Recommended)

This is the most common method. By prefixing the command with the variable assignment and using `sudo -E`, the variable is passed to the script's environment.

```
ssh -t user@remote-host "export ROOT_PASSWORD='your-secure-password'; sudo -E
/path/to/root_audit.sh --mode remediate"
```

Note: The `-t` flag is often required for `sudo` prompts over SSH.

2. Piped Execution (No script on host)

If the script is not already present on the remote host, you can stream it over SSH while passing the environment.

```
cat root_audit.sh | ssh user@remote-host "export ROOT_PASSWORD='your-secure-
password'; sudo -E bash -s -- --mode remediate"
```

3. SSH Configuration (SendEnv)

If your organization allows it, you can configure SSH to pass the environment variable automatically.

1. **Remote Host**: Add `AcceptEnv ROOT_PASSWORD` to `/etc/ssh/sshd_config`.
2. **Local Host**: Use `SendEnv` in your SSH command.

```
export ROOT_PASSWORD='your-secure-password'
ssh -o SendEnv=ROOT_PASSWORD user@remote-host "sudo -E /path/to/root_audit.sh --
mode remediate"
```

Simulation Mode

The `--simulate` flag allows you to dry-run the remediation process. It performs a full audit and sudo safety check, then explains every step it *would* take without modifying the system.

Example Simulation

```
sudo ./root_audit.sh --mode remediate --generate --simulate
```

Simulation Output Breakdown

- **Password**: Explains that a 32-character password would be enforced.
- **Locking**: Explains that the account would be locked in the shadow file.
- **SSH Hardening**: Details the atomic modification of `sshd_config` and the service reload.
- **Safety**: Confirms that backups and immutable attribute handling would be performed.

JSON Mode for CI/CD

The `--json` flag enables machine-readable output on `STDOUT`. This is ideal for integration with automation tools, monitoring systems, and security orchestrators.

Example Usage

```
sudo ./root_audit.sh --mode audit --json
```

JSON Schema

The output is a structured object containing:

- `timestamp`: ISO-8601 execution time.
- `target_user`: Target account name.
- `mode`: Execution mode (`audit` or `remediate`).
- `immutable`: Boolean indicating if `chattr +i` was detected on scoped files.
- `status`: `success` or `failure`.
- `audit`: Granular results of all security checks (Audit mode only).
- `remediation`: Summary of actions taken (Remediate mode only).

`jq` Support: If `jq` is installed, the output will be pretty-printed. If not, the script falls back to raw JSON strings. Standard log messages are preserved on `STDERR`.

7. Simulation with JSON Output

Combine flags to preview remediation steps in a machine-readable format without making system changes.

```
sudo ./root_audit.sh --mode remediate --generate --simulate --json
```

This is the recommended pattern for validating remediation logic in CI/CD pipelines before production deployment.

JSON Output Examples

Audit Mode

```
{
  "timestamp": "2026-04-23T14:47:54-05:00",
  "target_user": "root",
  "mode": "audit",
  "immutable": false,
  "simulate": false,
  "status": "failure",
  "audit": {
    "password_exists": "true",
    "account_locked": "false",
    "ssh_disabled": "false",
    "hash_strong": "true"
  }
}
```

Remediate Mode

```
{
  "timestamp": "2026-04-23T14:50:12-05:00",
  "target_user": "root",
  "mode": "remediate",
  "immutable": true,
  "simulate": false,
  "status": "success",
  "remediation": {
    "password_updated": true,
    "account_locked": true,
    "ssh_hardened": true,
    "backup_created": true,
    "sudo_safety_passed": "true"
  }
}
```